

One-Dimensional Fatigue Modeling of Pure Aluminum: Deterministic Calibration and a Probabilistic Spacing Hypothesis

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Abstract

We derive one intrinsic conservative energy for a reduced single-crystal fatigue state with normal layer spacing a and one crystallographic registry s . A row–row interaction $W(d, s)$ is only a kernel: the active local lattice potential is the multiplicity-free sum $U_0(a, s) = \sum_{k \geq 1} W(ka, s)$, with the same collective s at every layer. Mellin transformation and Poisson summation yield an exact Bessel–Lambert representation of the new two-index lattice sum H_q . For the 12–6 case its half-integer Bessel terms reduce exactly to verified polylogarithms. Normal opening and slip are then an identity decomposition of the same U_0 , so they are not double counted. Uniaxial tensile stress enters only as generalized driving forces in a two-coordinate Smoluchowski equation. The four governing observables are mean spacing, mean intrinsic energy, cumulative irreversible dissipation, and normalization/survival. Plasticity requires a residual transition of the unwrapped slip-well index; crack initiation is first passage through the negative-curvature normal barrier, not an empirical energy threshold. The present numerical examples are dimensionless mechanism tests, not calibrated aluminum-life predictions.

1 Scope and fatigue-crack-initiation objective

The physical target is repeated uniaxial normal tension of a pure single crystal. The microscopic state uses one normal spacing a and one scalar registry s for a declared slip system. The probability state space is thus two-dimensional, but this does not introduce a two-dimensional continuum stress criterion. The only applied waveform is uniaxial. Generalized Lennard–Jones energetics are retained deliberately so that the landscape is derived rather than looked up. EAM and DFT are future quantitative checks, not governing potentials in this theory.

2 Definitions and counting convention

Consider one reference row and rows at normal distances $a, 2a, 3a, \dots$. Within every row the tangential repeat is b . The scalar s is one collective relative registry; it is neither multiplied by the normal-layer index nor independently redrawn for each layer. Once one reference layer has been chosen, its interaction with layer k occurs once. Consequently

$$U_0(a, s) = \sum_{k=1}^{\infty} W(ka, s), \tag{1}$$

with no multiplicity before W . In contrast, $\sum_k kW(ka, s)$ counts all cross-interface layer pairs of two half-spaces. That is a distinct interface energy and is not the local-state convention. Similarly $W(ka, ks)$ would assign a different registry to every layer and is excluded by the definition of the collective coordinate.

3 Generalized Lennard–Jones m – n pair potential

We use the well-depth convention

$$v_{m,n}(r) = C_{m,n}\varepsilon_{\text{LJ}} \left[\left(\frac{\sigma_{\text{LJ}}}{r} \right)^m - \left(\frac{\sigma_{\text{LJ}}}{r} \right)^n \right], \quad C_{m,n} = \frac{m}{m-n} \left(\frac{m}{n} \right)^{n/(m-n)}. \quad (2)$$

It gives $\min v = -\varepsilon_{\text{LJ}}$ at $r_e = \sigma_{\text{LJ}}(m/n)^{1/(m-n)}$. For $m = 12, n = 6$, $C_{12,6} = 4$, so the usual $4\varepsilon_{\text{LJ}}$ convention is recovered rather than silently mixed with a different coefficient definition. The full multilayer absolute energy below requires

$$m > n > 2. \quad (3)$$

The weaker condition $q > 1$ suffices for an individual row kernel.

4 Single-row kernel $W(d, s)$

For row separation d and site index $p \in \mathbb{Z}$,

$$W_{m,n}(d, s) = \sum_{p=-\infty}^{\infty} v_{m,n}(\sqrt{d^2 + (pb + s)^2}). \quad (4)$$

Put $\delta = s/b$, $\eta = d/b$ and, for $q > 1$,

$$B_q(\delta, \eta) = \sum_{p=-\infty}^{\infty} [(p + \delta)^2 + \eta^2]^{-q/2}. \quad (5)$$

Then

$$W_{m,n} = C_{m,n}\varepsilon_{\text{LJ}} [(\sigma_{\text{LJ}}/b)^m B_m - (\sigma_{\text{LJ}}/b)^n B_n]. \quad (6)$$

5 Exact Mellin–Poisson–Bessel representation of W

For $x > 0$ and $\text{Re } \alpha > 0$, the Gamma integral gives

$$x^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty t^{\alpha-1} e^{-xt} dt. \quad (7)$$

Applying it termwise to Eq. (5) is justified by Tonelli’s theorem for real $q > 1$. Shifted Gaussian Poisson summation gives

$$\sum_{p \in \mathbb{Z}} e^{-t(p+\delta)^2} = \sqrt{\frac{\pi}{t}} \sum_{\ell \in \mathbb{Z}} e^{-\pi^2 \ell^2 / t} e^{2\pi i \ell \delta}. \quad (8)$$

The required nonzero reciprocal integral follows from the defining identity for the modified Bessel function of the second kind [3],

$$K_\nu(x) = \frac{1}{2} \int_0^\infty u^{\nu-1} \exp\left[-\frac{x}{2}(u + u^{-1})\right] du, \quad (9)$$

because

$$\int_0^\infty t^{\nu-1} e^{-\beta t - \gamma/t} dt = 2(\gamma/\beta)^{\nu/2} K_\nu(2\sqrt{\beta\gamma}). \quad (10)$$

Separating the $\ell = 0$ mode and combining $\pm\ell$ yields the exact identity

$$B_q(\delta, \eta) = \frac{\sqrt{\pi}}{\Gamma(q/2)} \left[\Gamma\left(\frac{q-1}{2}\right) \eta^{1-q} + 4 \sum_{\ell=1}^\infty \cos(2\pi\ell\delta) \left(\frac{\pi\ell}{\eta}\right)^{(q-1)/2} K_{(q-1)/2}(2\pi\ell\eta) \right]. \quad (11)$$

No small-slip, harmonic, or Taylor approximation was made.

6 Multi-layer intrinsic lattice potential

Now redefine $\eta = a/b$ and introduce

$$H_q(\delta, \eta) = \sum_{k=1}^\infty B_q(\delta, k\eta) = \sum_{k=1}^\infty \sum_{p=-\infty}^\infty [(p + \delta)^2 + k^2\eta^2]^{-q/2}. \quad (12)$$

This two-index absolute sum converges for $q > 2$, as follows by comparison with the radial integral of R^{-q} over a two-dimensional lattice. The intrinsic potential is therefore

$$\begin{aligned} U_0(a, s) &= \sum_{k=1}^\infty \sum_{p=-\infty}^\infty v_{m,n} \left(\sqrt{k^2 a^2 + (pb + s)^2} \right) \\ &= C_{m,n} \varepsilon_{\text{LJ}} [(\sigma_{\text{LJ}}/b)^m H_m(\delta, \eta) - (\sigma_{\text{LJ}}/b)^n H_n(\delta, \eta)]. \end{aligned} \quad (13)$$

It depends only on atomic positions. External stress work is absent.

7 Exact H_q series

Let $\nu_q = (q-1)/2$. The zero Fourier modes sum as $\sum_{k \geq 1} (k\eta)^{1-q} = \eta^{1-q} \zeta(q-1)$. The nonzero modes give the Bessel–Lambert series

$$\mathcal{K}_\nu(x) = \sum_{k=1}^\infty k^{-\nu} K_\nu(kx). \quad (14)$$

Hence

$$H_q = \frac{\sqrt{\pi}}{\Gamma(q/2)} \left[\Gamma(\nu_q) \eta^{1-q} \zeta(q-1) + 4 \sum_{\ell=1}^\infty \cos(2\pi\ell\delta) \left(\frac{\pi\ell}{\eta}\right)^{\nu_q} \mathcal{K}_{\nu_q}(2\pi\ell\eta) \right]. \quad (15)$$

Absolute convergence permits the reordering. Exponential decay of K_ν makes the reciprocal form rapidly convergent for $\eta > 0$; its truncations are numerical controls, not physical cutoffs.

8 Normal/slip decomposition from the same U_0

For one reference (a_0, s_0) , add and subtract $U_0(a, s_0)$:

$$\boxed{\begin{aligned}\Delta U_0(a, s) &= U_0(a, s) - U_0(a_0, s_0) \\ &= \underbrace{U_0(a, s_0) - U_0(a_0, s_0)}_{\Delta U_n(a)} + \underbrace{U_0(a, s) - U_0(a, s_0)}_{V_{\text{slip}}(a, s)}.\end{aligned}} \quad (16)$$

This is an exact identity. It replaces the former practice of treating a collinear U_∞ and a two-row W as independent candidates. Those reduced geometries may be archived, but are not added to form the present fundamental energy.

9 Slip-excess ΔH_q

With $\delta_0 = s_0/b$, the zero mode cancels exactly:

$$\boxed{\begin{aligned}\Delta H_q &= \frac{4\sqrt{\pi}}{\Gamma(q/2)} \sum_{\ell=1}^{\infty} [\cos(2\pi\ell\delta) - \cos(2\pi\ell\delta_0)] \left(\frac{\pi\ell}{\eta}\right)^{\nu_q} \\ &\quad \times \sum_{k=1}^{\infty} k^{-\nu_q} K_{\nu_q}(2\pi\ell k\eta).\end{aligned}} \quad (17)$$

Then

$$V_{\text{slip}} = C_{m,n} \varepsilon_{\text{LJ}} [(\sigma_{\text{LJ}}/b)^m \Delta H_m - (\sigma_{\text{LJ}}/b)^n \Delta H_n]. \quad (18)$$

At $\delta = \delta_0$ every bracket is zero, proving $V_{\text{slip}}(a, s_0) = 0$. Replacing δ by $\delta + 1$ leaves every cosine unchanged, proving period b . If $s_0 = 0$, $\cos(2\pi\ell\delta) - 1 = -2\sin^2(\pi\ell\delta)$.

10 LJ 12–6 polylogarithm special case

The independently expanded half-integer functions are

$$K_{5/2}(x) = \sqrt{\pi/(2x)} e^{-x} (1 + 3x^{-1} + 3x^{-2}), \quad (19)$$

$$K_{11/2}(x) = \sqrt{\pi/(2x)} e^{-x} (1 + 15x^{-1} + 105x^{-2} + 420x^{-3} + 945x^{-4} + 945x^{-5}). \quad (20)$$

In $k^{-\nu} K_\nu(kx)$, the square-root adds $k^{-1/2}$. Using $\sum_{k \geq 1} e^{-kx}/k^r = \text{Li}_r(e^{-x})$ gives

$$\mathcal{K}_{5/2}(x) = \sqrt{\pi/(2x)} [\text{Li}_3(e^{-x}) + 3x^{-1} \text{Li}_4(e^{-x}) + 3x^{-2} \text{Li}_5(e^{-x})], \quad (21)$$

$$\begin{aligned}\mathcal{K}_{11/2}(x) &= \sqrt{\pi/(2x)} [\text{Li}_6(e^{-x}) + 15x^{-1} \text{Li}_7(e^{-x}) + 105x^{-2} \text{Li}_8(e^{-x}) \\ &\quad + 420x^{-3} \text{Li}_9(e^{-x}) + 945x^{-4} \text{Li}_{10}(e^{-x}) + 945x^{-5} \text{Li}_{11}(e^{-x})].\end{aligned} \quad (22)$$

Substitution into Eq. (15) is the explicit polylog form. The coefficients were generated from the exact half-integer formula and checked against both unsimplified Bessel sums and real-space double sums.

11 Unwrapped slip and plastic deformation

The periodic energy does not justify folding the state history. Write

$$\boxed{s = s_0 + zb + \tilde{s}, \quad z \in \mathbb{Z},} \quad (23)$$

and let $\mathcal{W}_z$ be well z . Define

$$p_z = \iint_{\mathcal{W}_z} P \, da \, ds, \quad qquad \langle z \rangle = \sum_z z p_z. \quad (24)$$

For an independently specified homogenization thickness,

$$\boxed{\gamma_p = \frac{b}{h_{\text{slip}}} \langle z \rangle, \quad qquad \varepsilon_p = M \frac{b}{h_{\text{slip}}} \langle z \rangle.} \quad (25)$$

Intrawell lag is not plasticity. The reduced operational criterion is a residual $\Delta \langle z \rangle \neq 0$ after unloading and relaxation.

12 Tensile-driving probability dynamics

The only applied load is

$$\boxed{\sigma(t) = \sigma_m + \sigma_a \sin \omega t, \quad Q_a = A_0 \sigma(t), \quad Q_s = A_0 M \sigma(t).} \quad (26)$$

$M = (\mathbf{e} \cdot \mathbf{n})(\mathbf{e} \cdot \mathbf{d})$ is the signed resolved-slip projection for one declared slip plane and direction. There is no independent shear fatigue input.

Eliminating isothermal bath degrees of freedom after velocity relaxation gives $\dot{q}_i = -M_i(\partial_{q_i} U_0 - Q_i) + \sqrt{2k_B T M_i} \xi_i$. The fluctuation–dissipation coefficient is fixed, and the Fokker–Planck equation is [1, 2]

$$\boxed{\partial_t P = -\partial_a J_a - \partial_s J_s,} \quad (27)$$

$$\boxed{J_a = -M_a[P(\partial_a U_0 - Q_a) + k_B T \partial_a P], \quad J_s = -M_s[P(\partial_s U_0 - Q_s) + k_B T \partial_s P].} \quad (28)$$

This PDE is the underlying evolution law; it is not counted as one of the four observable equations below.

13 The four governing equations

For a closed bonded domain Ω_b , the named equations are

$$\text{G1: } \boxed{\bar{a}(t) = \iint_{\Omega_b} a P \, da \, ds,} \quad (29)$$

$$\text{G2: } \boxed{\bar{U}(t) = \iint_{\Omega_b} [U_0(a, s) - U_0(a_0, s_0)] P \, da \, ds,} \quad (30)$$

$$\text{G3: } \boxed{E_{\text{hyst}}(t) = \int_0^t \dot{D}_{\text{irr}}(t') \, dt', \quad \dot{D}_{\text{irr}} = \iint_{\Omega_b} \left[\frac{J_a^2}{M_a P} + \frac{J_s^2}{M_s P} \right] da \, ds \geq 0,} \quad (31)$$

$$\text{G4: } \boxed{\iint_{\Omega_b} P(a, s, t) \, da \, ds = 1.} \quad (32)$$

With no-flux boundaries, integration by parts gives $d\bar{a}/dt = \iint J_a \, da \, ds$ and

$$\frac{d\bar{U}}{dt} = \iint (\partial_a U_0 J_a + \partial_s U_0 J_s) \, da \, ds. \quad (33)$$

The omitted boundary terms must be restored for absorption. Because $U_0(a, s + b) = U_0(a, s)$, an equivalent-well transition can lower $\bar{U}$; therefore $\bar{U}$ is not a cumulative damage variable.

14 Hysteresis and irreversible energy accumulation

The nonequilibrium free energy

$$\mathcal{F}[P, t] = \iint P[U_0 - Q_a a - Q_s s + k_B T \ln(P/P_*)] \, da \, ds \quad (34)$$

satisfies, for no-flux or periodic boundaries,

$$\frac{d\mathcal{F}}{dt} = -\dot{Q}_a \bar{a} - \dot{Q}_s \bar{s} - \dot{D}_{\text{irr}}. \quad (35)$$

Thus finite mobility produces phase lag because current needs finite time to redistribute P ; diffusion alone is not the cause. In a quasistatic closed cycle the conditional distribution returns to the same state and loop area tends to zero. For cycle N ,

$$\Delta E_{\text{hyst}, N} = \int_{(N-1)T_c}^{NT_c} \dot{D}_{\text{irr}} \, dt, \quad \text{quad} E_{\text{hyst}}^{(N)} = \sum_{j=1}^N \Delta E_{\text{hyst}, j}. \quad (36)$$

E_{hyst} is total irreversible/hysteretic dissipation. It may include heat transferred to eliminated modes as well as microstructural irreversible work, so it is tracked as history but not equated automatically with stored damage energy.

15 Crack-initiation absorbing boundary and first passage

At fixed s, t , stationary normal states solve the full-energy force balance

$$\boxed{\partial_a U_0(a^\dagger, s) = Q_a(t), \quad \partial_a^2 U_0(a^\dagger, s) < 0.} \quad (37)$$

The positive-curvature root is the stable bonded state; the outer negative-curvature root defines the absorbing barrier. With absorption,

$$\boxed{S(t) = \iint_{\Omega_b(t)} P \, da \, ds \leq 1, \quad \text{quad} P_{\text{crack}}(t) = 1 - S(t).} \quad (38)$$

If the moving boundary is a graph $a = a^\dagger(s, t)$ and registry edges are periodic or no-flux, Reynolds transport gives the relative outflux

$$\boxed{-\dot{S} = \int \left[J_a - J_s \partial_s a^\dagger - P \partial_t a^\dagger \right]_{a=a^\dagger} \, ds.} \quad (39)$$

For an ideal absorbing boundary $P = 0$, its motion term vanishes. A finite-grid radiation implementation must retain it. Neither an arbitrary distance nor $E_{\text{hyst}} > E_c$ defines fracture.

16 Numerical consistency checks

The implementation evaluates direct (k, p) sums, Fourier–Bessel sums, Bessel–Lambert sums, and the 12–6 polylog closure independently. Table 1 records representative double-precision results; direct sums use $k_{\max} = 400, p_{\max} = 800$ and reciprocal sums use 20 modes. The polylog and

q	(δ, η)	direct H_q	Bessel–Lambert H_q	relative error
6	(0,0.7)	9.39075449597613	9.39075449604469	7.30×10^{-12}
6	(0.23,1)	1.23234977228797	1.23234977229983	9.63×10^{-12}
6	(0.5,1.4)	0.224780520495904	0.224780520498402	1.11×10^{-11}
12	(0,0.7)	72.4515258960140	72.4515258960140	1.96×10^{-16}
12	(0.23,1)	0.799764218140630	0.799764218140630	1.39×10^{-16}
12	(0.5,1.4)	0.0175417129797649	0.0175417129797649	7.91×10^{-16}

Table 1: Independent direct and analytical evaluations.

unsimplified Bessel–Lambert values agree to at most 4×10^{-15} relative tolerance in the automated tests. Periodicity, even symmetry, $V_{\text{slip}}(s_0) = 0$, analytical a and s derivatives, unweighted layer counting, probability conservation, nonnegative dissipation, and grid/time refinement are separately tested. Units are $[U_0] = \text{J}$, $[\partial_i U_0] = \text{N}$, $[P] = \text{m}^{-2}$, $[J_i] = \text{m}^{-1}\text{s}^{-1}$, $[\dot{D}_{\text{irr}}] = \text{J s}^{-1}$, and $[E_{\text{hyst}}] = \text{J}$.

17 Model limitations and future quantitative extension

The exact sums do not determine the representative area A_0 , mobilities, memory time, active slip system, or h_{slip} . One collective registry does not contain dislocation multiplication, hardening, backstress, or multiple-slip interactions. These unresolved quantities prevent a quantitative aluminum fatigue-life claim. A future FCC/EAM/DFT calculation may validate or extend the generalized-LJ landscape, but it does not replace the current analytical governing energy [5]. Likewise, a CAD/FEM mesh supplies geometry and scalar normal stress only; it does not add shear or multiaxial constitutive physics.

18 Complete symbol and computational index

Symbol	Physical meaning	Unit	Definition, evaluation, and location
a, a_0	spacing and reference spacing	m	State coordinate and reference in Sec. 8.
$a^{\ddagger}$	outer normal saddle	m	Negative-curvature root of Eq. (37); bracket the full derivative.
b	row repeat/slip period	m	Declared crystallographic repeat in Sec. 4.
$s, s_0, \tilde{s}$	unwrapped, reference, and intrawell registry	m	$s = s_0 + zb + \tilde{s}$, Sec. 11.
z	unwrapped well index	1	Integer basin label; compute before folding s .

Symbol	Physical meaning	Unit	Definition, evaluation, and location
m, n, q	LJ and generic exponents	1	$m > n > 2$ for U_0 ; $q > 1$ suffices for one row.
$\varepsilon_{\text{LJ}}, \sigma_{\text{LJ}}$	pair well depth and zero crossing	J, m	Independent pair parameters in Eq. (2).
$C_{m,n}$	well-depth normalization	1	Closed expression in Eq. (2).
p, k, ℓ	row-site, normal-layer, reciprocal indices	1	$p \in \mathbb{Z}$, $k, \ell \geq 1$; Secs. 4–6.
δ, δ_0, η	normalized registries and spacing	1	$s/b, s_0/b, a/b$.
B_q	one-row inverse-power kernel	1	Direct Eq. (5); Bessel Eq. (11).
H_q	unweighted multilayer sum	1	Direct Eq. (12); exact series Eq. (15).
ΔH_q	registry excess of H_q	1	Cosine-difference series Eq. (17).
K_ν	modified Bessel function, second kind	1	Integral definition Eq. (9); standard special-function evaluation.
$\mathcal{K}_\nu$	Bessel–Lambert series	1	$\sum k^{-\nu} K_\nu(kx)$ in Eq. (14).
Li_r	polylogarithm	1	$\sum e^{-kx}/k^r$; 12–6 closure in Sec. 10.
$v_{m,n}$	pair potential	J	Eq. (2); pair counted once.
$W(d, s)$	one row–row kernel	J	Direct site sum Eq. (4); not the final lattice energy.
U_0	intrinsic multilayer potential	J	Unweighted $\sum_k W(ka, s)$, Eqs. (1) and (13); excludes external work.
$\Delta U_0, \Delta U_n$	total excess, normal	J	Exact common-energy identity
V_{slip}	part, slip part		Eq. (16).
P	spacing–registry density	m^{-2}	Underlying continuity law Eq. (27).
J_a, J_s	probability currents	$\text{m}^{-1}\text{s}^{-1}$	Force/diffusion currents in Eq. (28).
M_a, M_s	constant mobilities	$\text{m}^2\text{J}^{-1}\text{s}^{-1}$	Long-time overdamped coefficients; Einstein diffusion is $k_B T M_i$.
Q_a, Q_s	tensile-derived generalized forces	N	$A_0\sigma$ and $A_0 M\sigma$, Eq. (26).
M	signed Schmid factor	1	$(\mathbf{e} \cdot \mathbf{n})(\mathbf{e} \cdot \mathbf{d})$ for one declared slip system.
A_0	representative mechanical area	m^2	Converts stress to generalized force; not a FEM or correlation area.
T, k_B	bath temperature and Boltzmann constant	K, J K $^{-1}$	Isothermal fluctuation–dissipation assumption.
$\bar{a}, \bar{U}$	G1 mean spacing and G2 mean intrinsic excess	m, J	Eqs. (29)–(30).
$\dot{D}_{\text{irr}}, E_{\text{hyst}}$	dissipation rate and cumulative hysteresis	J s $^{-1}$, J	G3, Eq. (31); not automatically stored damage.
S, P_{crack}	survival and initiation probability	1	G4 extension, Eq. (38).

Symbol	Physical meaning	Unit	Definition, evaluation, and location
h_{slip}	slip homogenization thickness	m	Independently specified mapping scale, Eq. (25).
γ_p, ε_p	plastic shear and tensile strain	1	Derived from residual $\langle z \rangle$, Eq. (25).
Ω_b	bonded state domain	m ²	Reflecting or absorbing domain in Sec. 15.
$\omega, \sigma_m, \sigma_a$	angular frequency, mean and amplitude	s ⁻¹ , Pa, Pa	Uniaxial waveform Eq. (26).

한국어 연구

이 연구의 활성 위치에너지는 단일 row kernel W 자체가 아니라

$$U_0(a, s) = \sum_{k=1}^{\infty} W(ka, s)$$

이다. 기준 layer와 $a, 2a, 3a, \dots$ 떨어진 각 layer의 상호작용을 한 번씩 세므로 앞에 multiplicity k 가 없다. $\sum kW(ka, s)$ 는 두 half-space 사이의 모든 layer pair를 세는 별도 interface convention이다. 또한 s 는 하나의 collective/unwrapped slip 좌표이므로 모든 k 에 동일하며 ks 나 js 를 쓰지 않는다.

B_q 의 기존 Mellin–Poisson–Bessel 유도를 보존한 채 다시 k 에 대해 합하면 식 (15)의 정확한 H_q 를 얻는다. 이는 Taylor 또는 harmonic 근사가 아니다. 전체 absolute double sum 때문에 $m > n > 2$ 가 필요하고, 단일 row에는 $q > 1$ 이면 분하다. 12-에서는 $K_{5/2}$ 와 $K_{11/2}$ 가 유한다항식으로 닫 식 (21)–(22)의 polylog 식이 된다. 계수는 직접 이중합과 Bessel 합을 서로 비교하여 검산다.

normal opening과 slip은 서로 다른 에너지를 더한 것이 아니다. 동일한 U_0 에 $U_0(a, s_0)$ 를 더하고 서 식 (16)처럼 ΔU_n 과 V_{slip} 으로 정확히 분해한다. 외부응력의 일은 intrinsic U_0 에 넣지 않고, 단축 인장응력 하나로부터 $Q_a = A_0\sigma$ 와 $Q_s = A_0M\sigma$ 만 확률방정식에 넣는다. 독립 shear 피로하중은 없다.

공식적인 4 governing equations는 G1 평균 층간거리, G2 평균 intrinsic 초과에너지, G3 누적 비가역 dissipation E_{hyst} , G4 정규화이다. U_0 가 s 에 대해 주기적이므로 slip 후 평균 lattice energy는 다시 아질 수 있다. 반면 $E_{\text{hyst}} = \int \dot{D}_{\text{irr}} dt$ 는 단조 비감소하지만, 그 전체가 재료에 저장된 damage energy라고 해하지 않는다.

소성은 단순한 phase lag가 아니다. $s = s_0 + zb + \tilde{s}$ 로 펼쳐 저장하고, unloading과 relaxation 뒤에도 $\Delta\langle z \rangle \neq 0$ 일 때만 소성변형으로 정의한다. 균열개시는 임의 a_c 또는 E_c 가 아니라 $\partial_a U_0 = Q_a$, $\partial_a^2 U_0 < 0$ 인 외 metastable barrier를 통과하는 확률유출로 정의한다. 경계가 직이면 식 (39)의 상대 flu를 사용한다.

현재 결과는 수학적/무차원 mechanism 검증이다. A_0 , mobility, memory, h_{slip} , dislocation hardening이 독립적으로 확정되기 전에는 알루미늄 수명이라고 주장하지 않는다. EAM/DFT는 향후 정량 검증용일 현재 fundamental potential을 대체하지 않는다.

기 및 계산

기호	물리적 의미	단위	정의/계산법/위치
$a, a_0, a^\ddagger$	층간거리, 기준거리, 외 장벽	m	$a^\ddagger$ 는 식 (37)의 음의 곡률 근.
b	row 반복거리/slip 주기	m	선택한 결정학적 반복거리.
$s, s_0, \tilde{s}, z$	펼친 slip, 기준, well 내부 좌표, well 지수	m, m, m, 1	$s = s_0 + zb + \tilde{s}$ Sec. 11.
m, n, q	LJ 지수와 일반 지수	1	전체 U_0 는 $m > n > 2$ 단일 row는 $q > 1$.
$\varepsilon_{\text{LJ}}, \sigma_{\text{LJ}}, C_{m,n}$	pair 우물깊이, 영점거리, 정규화계수	J, m, 1	식 (2) 피로 loop에 fitting하지 않는다.
p, k, ℓ	row site, normal layer, 역격자 지수	1	$p \in \mathbb{Z}$, $k, \ell \geq 1$ k 는 multiplicity가 아니다.
δ, δ_0, η	무차원 registry와 거리	1	$s/b, s_0/b, a/b$.
$B_q, H_q, \Delta H_q$	단일-row, 다층, slip-excess 합	1	식 (5), (15), (17).

기호	물리적 의미	단위	정의/계산법/위치
$K_\nu, \mathcal{K}_\nu, \text{Li}_r$	Bessel, Bessel-Lambert, polylog 함수	1	Secs. 5, 1.
$v_{m,n}, W$	pair potential과 row-row kernel	J	식 (2), (4) W 는 최종 에너지가 아니다.
U_0	intrinsic 다층 위치에너지	J	$\sum_{k \geq 1} W(ka, s)$ 외력 일을 포함하지 않는다.
$\Delta U_0, \Delta U_n$ V_{slip}	전체/normal/slip 초과에 너지	J	동일한 U_0 의 항등분해 식 (16).
P	(a, s) 확률밀도	m^{-2}	식 (27)로 진화한다.
J_a, J_s	확률 flu	$\text{m}^{-1}\text{s}^{-1}$	식 (28).
M_a, M_s	mobility	$\text{m}^2\text{J}^{-1}\text{s}^{-1}$	확산은 $k_B T M_i$ 로 고정된다.
Q_a, Q_s	인장에서 유도된 일반화 힘		$A_0\sigma, A_0M\sigma$ 독립 shear 입력은 없다.
M, A_0	signed Schmid factor, 대표 기계면적	$1, \text{m}^2$	A_0 는 FEM 요소면적이나 상관면적이 아니다.
$\bar{a}, \bar{U}$	G1 평균거리, G2 평균 intrinsic 에너지	m, J	식 (29), (30).
$\dot{D}_{\text{irr}}, E_{\text{hyst}}$	비가역률과 누적 hysteresis	$\text{J s}^{-1}, \text{J}$	G3 식 (31) 저장 damage와 동일시하지 않는다.
S, P_{crack}	생존/균열개시 확률	1	G4 확장 식 (38).
$h_{\text{slip}}, \gamma_p, \varepsilon_p$	균질화 두께, 소성 shear/ 인장 strain	$\text{m}, 1, 1$	$\langle z \rangle$ 로 식 (25) 계산.
T, k_B, Ω_b	bath 온도, Boltzmann 상수, 결합영역	$\text{K}, \text{J K}^{-1}, \text{m}^2$	확률동역학과 absorbing 경계에서 사용.
$\omega, \sigma_m, \sigma_a$	각주파수, 평균응력, 진	$\text{s}^{-1}, \text{Pa}, \text{Pa}$	단축 인장파형 식 (26).

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